Understand, try, resume

1.1 Rank, Value and Update

First use: S2 and S4.

Definition · A sequence is not a finished list

The index n means an integer; u_n is a real. The point of representation is $(n; u_n)$. One line connecting points can help reading but does not define terms to non-integer indices.

Solved Example · Two Similar Expressions

For $u_0 = 2$ and $u_{n+1} = 3u_n - 1$, we get $u_1 = 5$, $u_2 = 14$, $u_3 = 41$. Thus $u_2 + 1 = 15$ while that $u_{2+1} = 41$. In a loop, the variable `u` contains first u_n ; education `u = 3*u-1` replace it by u_{n+1}.

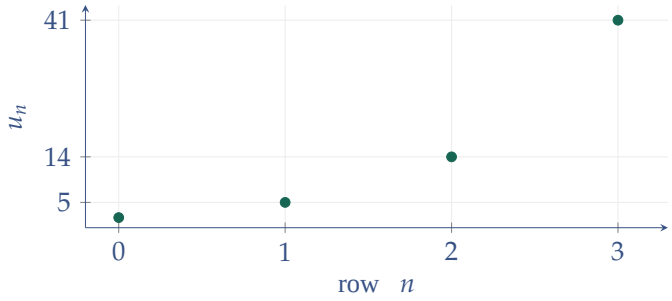


Figure 1.1. Read the term rank 2: his order is 14, not 2.

A1 · Immediate application For $v_0 = 1$, $v_{n+1} = 2v_n + 1$, build a table of rows 0 to 3, then place the four points. Calculate separately $v_2 + 1$ and v_{2+1}.

A2 · Partial guidance A loop must be calculated $u_0 + u_1 + u_2 + u_3$ for the rest of the example. of u=2; total=2 . How many updates do you need? Write the order of instructions and explain why adding before the update would count twice u_0.

1.2 A local hypothesis, a universal conclusion

First use: S5 and S7.

Example resolved · Recurrence of inequality

Let $w_0 = 1$ and $w_{n+1} = \frac{1}{2}w_n + 2$. Prove $P(n) : 1 \leq w_n < 4$. Initially $w_0 = 1 \in [1; 4[$. Fixes $n \in \mathbb{N}$ and suppose $P(n)$. Since $\frac{1}{2} > 0$, $\frac{1}{2} \leq \frac{1}{2}w_n < 2$, so $\frac{5}{2} \leq w_{n+1} < 4$. Like $\frac{5}{2} \geq 1$, this implies $P(n+1)$. The principle of recurrence provides guidance at all levels.

A3 · Proof to be completed For $a_0 = 2$, $a_{n+1} = 2a_n + 1$, complete: " $P(n)$ means $a_n = 3 \times 2^n - 1$. At range 0, ... Fixes . . . and suppose . . . . Then $a_{n+1} = \ldots = 3 \times 2^{n+1} - 1$. Therefore . . . "

A4 · Repair Evidence A student writes: "If $u_n \geq 2$, then $u_{n+1} = 2u_n \geq 4 \geq 2$. So all the terms are at least 2.' He did not give u_0. Build a counter-example, then indicate which given makes his proof complete.

1.3 Count before summation

First use: S10.

Property · Number of terms

For integers $p \leq n$, the sum $\sum_{k=p}^{n} u_k$ contains $n - p + 1$ terms. It can be verified in Writing first and last, before any formula.

Example resolved · A offset geometric sum

In $\sum_{k=3}^{7} 2^k$, there are five terms and the first is worth 8. The sum is 8 $\frac{1-2^5}{1-2} = 248$. For Demonstrate the formula without memorizing it in isolation, write $S = 1 + q + \cdots + q^n$, then subtract $qS = q + \cdots + q^{n+1}$: $(1-q)S = 1 - q^{n+1}$. The division presupposes $q \neq 1$.

A5 · Independent evidence Demonstrate by induction, for everything $n \in \mathbb{N}$, $\sum_{k=0}^{n}(2k+1) = (n+1)^2$. Explain why the term added in heredity is 2 $n + 3$, and not $2n + 1$.

1.4 Recognize a structure without confounding it with a boundary

First use: S11 and S13.

Example solved · Choose the offset

For $w_{n+1} = \frac{1}{2}w_n + 2$, the fixed value is 4. By subtracting 4 of the two members, $w_{n+1} - 4 = \frac{1}{2}(w_n - 4)$. Thus $w_n = 4 - 3\,2^{-n}$. The gap is not arbitrary: it makes the term disappear This explicit calculation does not yet imply a convergence theorem.

A6 · Choice of strategy A subscription costs 40 euros in the first year, then its cost increases each year by 5 euros. Another costs 40 euros and then increases by 10 % per year. Write both to compare the first three years and calculate the cumulative costs over ten years. number of terms.

A first step: First name the nature of the two developments. The tenth year corresponds to the rank 9 if the first is row 0.

Wait a minute at the courthouse. Explain the difference between "for everything n, $u_n \leq 4$" and "there exists n such that $u_n \leq 4$". Give a follow-up that checks the second but not the first.

Review postponed · S20 and then S41 Take over $w_n = 4 - 3\,2^{-n}$ to prove its convergence; in probability, look again for the shift when a probability checks a relationship affine.

Research activity

One tank powered every day · 35 min.

A tank initially contains 100 litres of water. Every day, 20% of the present volume is taken, Then we add 30 liters. For all natural n, we note u_n the volume of water, in litres, after these two transactions n So, $u_0 = 100$.

1. Calculate u_1 and u_2, then express u_{n+1} according to the u_n.
2. What value of the volume would remain unchanged after a day? Demonstrate by induction that, for every $n \in \mathbb{N}$, $100 \leq u_n < 150$.

3. For every $n \in \mathbb{N}$, we're laying $v_n = u_n - 150$. Show that (v_n) is geometrical. terms of v_n and u_n according to the n.

4. Write a Python function without argument that returns the first integer n for which $u_n > 145$. Run the program and interpret the result.

5. One student calculated the first hundred terms and found that they were less than 150. Is it enough to show that all the terms are? Justify.